Project 02 — Multi-Site Expansion + DHCP

Enterprise Network Labs Series

Author: Leonel Chongong

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Platform: Cisco CML 2.9 | IOL + IOL-L2 + Dnsmasq

Series: Build → Verify → Break → Fix

Prerequisite: Project 01 — Campus Foundation must be complete

Time Estimate: 3–4 hours

Expand the HQ campus to a two-site enterprise with WAN connectivity, centralized DHCP via relay agent, IPv6 dual-stack, and DNS end-to-end.

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Introduction & Lab Requirements

Problem Statement

HQ exists as an isolated campus (Project 01). The business opens a second site (Branch). You need to: build the branch matching HQ's VLAN scheme, connect both sites over a point-to-point WAN link, deploy centralized DHCP serving all 8 VLANs across both sites via relay agent, add IPv6 dual-stack, and prove DNS resolution works end-to-end.

What You Will Build

- BR-RTR1 — Branch router (router-on-a-stick for 5 VLANs)
- BR-DSW1 — Branch distribution switch (STP root, trunk hub)
- BR-ASW1 — Branch access switch (endpoint connections)
- WAN /30 link between HQ-RTR1 (10.0.0.1) and BR-RTR1 (10.0.0.2)
- Static routing — explicit per-subnet routes both directions
- HQ-DHCP-DNS — Dnsmasq server serving 8 DHCP pools via ip helper-address relay
- IPv6 dual-stack: /126 WAN, /64 VLAN 100 at both sites, SLAAC
- DNS with A records for all infrastructure and endpoints

CCNA Relevance

- ip helper-address DHCP relay (3.x)
- Static routing (3.x)
- Point-to-point WAN design
- IPv6 unicast routing and SLAAC (1.x)
- Router-on-a-Stick (3.x — reinforced from Project 01)
- STP root election (2.x — reinforced)

Lab Requirements

Requirement
Project 01 complete and running
CML file: Enterprise-network-labs.yaml (adds branch nodes to existing topology)
Additional nodes: BR-RTR1 (IOL), BR-DSW1 (IOL-L2), BR-ASW1 (IOL-L2)
HQ-DHCP-DNS (Dnsmasq container), PC-BR1 (Alpine), PC-BR2 (Alpine)

IMPORTANT — Dnsmasq Node Config: Dnsmasq nodes in CML are configured via the CONFIG tab, not the console. You must paste TWO config files: the startup script (sets IP, starts dnsmasq) and dnsmasq.conf. Do NOT use bind-interfaces — it causes a container crash (exit code 30). Do NOT use interface=eth0 — it blocks relayed DHCP packets.

Network Topology & IP Addressing

Full Topology

```
HQ Site (10.1.x.x) Branch Site (10.2.x.x)

[HQ-RTR1]■■E0/1■■■
```

Interface	IPv6 Address	Purpose
HQ-RTR1 E0/1	2001:db8:0:1::1/126	WAN HQ-side
BR-RTR1 E0/1	2001:db8:0:1::2/126	WAN Branch-side
HQ-RTR1 E0/0.100	2001:db8:1:100::1/64	HQ Engineering IPv6 GW
BR-RTR1 E0/0.100	2001:db8:2:100::1/64	Branch Engineering IPv6 GW

DHCP Pool Design

Dnsmasq uses 'set' tags to select the correct pool based on the relay giaddr (the subinterface IP of the router that forwarded the DHCP request).

Pool Tag	Subnet	Range	Gateway	DNS	VLAN
hq-eng	10.1.100.0/24	.100–.200	10.1.100.1	10.1.99.50	100 HQ
hq-sales	10.1.200.0/24	.100–.200	10.1.200.1	10.1.99.50	200 HQ
hq-guest	10.1.44.0/24	.100–.200	10.1.44.1	10.1.99.50	300 HQ
hq-voice	10.1.50.0/24	.100–.200	10.1.50.1	10.1.99.50	500 HQ
br-eng	10.2.100.0/24	.100–.200	10.2.100.1	10.1.99.50	100 Branch
br-sales	10.2.200.0/24	.100–.200	10.2.200.1	10.1.99.50	200 Branch
br-guest	10.2.44.0/24	.100–.200	10.2.44.1	10.1.99.50	300 Branch
br-voice	10.2.50.0/24	.100–.200	10.2.50.1	10.1.99.50	500 Branch

Phase 1: Branch Site Base Configuration

Configure the full branch switching and routing infrastructure from scratch.

Step 1.1 — Configure BR-RTR1 (Router-on-a-Stick)

```
enable
configure terminal
hostname BR-RTR1
lldp run

interface Ethernet0/0
description TRUNK-TO-BR-DSW1-E0/0
no ip address
no shutdown

interface Ethernet0/0.100
description GATEWAY-VLAN100-ENGINEERING
encapsulation dot1Q 100
ip address 10.2.100.1 255.255.255.0
no shutdown

interface Ethernet0/0.200
description GATEWAY-VLAN200-SALES
encapsulation dot1Q 200
ip address 10.2.200.1 255.255.255.0
no shutdown

interface Ethernet0/0.300
description GATEWAY-VLAN300-GUEST
encapsulation dot1Q 300
ip address 10.2.44.1 255.255.255.0
no shutdown

interface Ethernet0/0.500
description GATEWAY-VLAN500-VOICE
encapsulation dot1Q 500
ip address 10.2.50.1 255.255.255.0
no shutdown
```

```
interface Ethernet0/0.999
description GATEWAY-VLAN999-MGMT
encapsulation dot1Q 999
ip address 10.2.99.1 255.255.255.0
no shutdown

interface Ethernet0/0.1000
description NATIVE-VLAN1000-UNUSED
encapsulation dot1Q 1000 native
no shutdown

end
write memory
```

WHY build Voice VLAN 500 now: avoids topology rework in Project 11 QoS. WHY native subinterface: explicitly tag native VLAN to prevent VLAN hopping.

Step 1.2 — Configure BR-DSW1 (Distribution Switch)

```
enable
configure terminal
hostname BR-DSW1
lldp run

ip routing

vlan 100
name Engineering
vlan 200
name Sales
vlan 300
name Guest
vlan 500
name Voice
vlan 999
name Management
vlan 1000
name NATIVE-UNUSED
```

```
interface vlan 999
ip address 10.2.99.2 255.255.255.0
no shutdown

ip default-gateway 10.2.99.1

spanning-tree mode rapid-pvst
spanning-tree vlan 100,200,300,500,999,1000 priority 4096

interface Ethernet0/0
description TRUNK-TO-BR-RTR1-E0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 100,200,300,500,999,1000
no shutdown

interface Ethernet0/1
description TRUNK-TO-BR-ASW1-E0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 100,200,300,500,999,1000
no shutdown

end
write memory
```

WHY ip routing enabled: Project 03 OSPF needs DSW1 as an area router. WHY priority 4096: guarantees BR-DSW1 wins root election for all VLANs.

Step 1.3 — Configure BR-ASW1 (Access Switch)

```
enable
configure terminal
hostname BR-ASW1
lldp run
```

```
ip routing
ip route 0.0.0.0 0.0.0.0 10.2.99.1

vlan 100
name Engineering
vlan 200
name Sales
vlan 300
name Guest
vlan 500
name Voice
vlan 999
name Management
vlan 1000
name NATIVE-UNUSED

interface vlan 999
ip address 10.2.99.3 255.255.255.0
no shutdown

spanning-tree mode rapid-pvst

interface Ethernet0/0
description TRUNK-TO-BR-DSW1-E0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 100,200,300,500,999,1000
no shutdown

interface Ethernet1/0
description ACCESS-PC-BR1-VLAN100
switchport mode access
switchport access vlan 100
switchport nonegotiate
spanning-tree portfast
```

```
 spanning-tree bpduguard enable
 no shutdown


 interface Ethernet1/1
 description ACCESS-PC-BR2-VLAN200
 switchport mode access
 switchport access vlan 200
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown


 end
 write memory
```

WHY ip routing instead of ip default-gateway: ip default-gateway cannot route replies to REMOTE subnet sources. When HQ-RTR1 (10.0.0.1) pings BR-ASW1, the reply must return to a remote subnet — this needs the routing engine.

> **IMPORTANT:** IOL-L2 Interface Naming: BR-ASW1 uses BOTH Ethernet0/x AND Ethernet1/x. Slot 0 connects to uplinks (trunks). Slot 1 connects to endpoints. Always verify with 'show cdp neighbors' after cabling.

Phase 1 Verification

```
 ! On BR-RTR1:
 show ip interface brief


 ! On BR-DSW1:
 show spanning-tree vlan 100
 show interfaces trunk
```

```
BR-RTR1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
Ethernet0/0            unassigned      YES TFTP   up                    up
Ethernet0/0.100        10.2.100.1      YES manual up                    up
Ethernet0/0.200        10.2.200.1      YES manual up                    up
Ethernet0/0.300        10.2.44.1       YES manual up                    up
Ethernet0/0.500        10.2.50.1       YES manual up                    up
Ethernet0/0.999        10.2.99.1       YES manual up                    up
Ethernet0/0.1000       unassigned      YES unset  up                    up
Ethernet0/1            unassigned      YES TFTP   administratively down down
Ethernet0/2            unassigned      YES TFTP   administratively down down
Ethernet0/3            unassigned      YES TFTP   administratively down down
BR-RTR1#
```

Figure 1.1 — BR-RTR1 interface brief confirming all subinterfaces up/up

Phase 2: WAN Connectivity & Static Routing

Objective: Connect HQ and Branch over a point-to-point /30 WAN link. Add static routes so both sites can reach all subnets.

WHY /30 subnet: A /30 gives exactly 2 usable host addresses — perfect for point-to-point links where only 2 routers connect.

WHY bandwidth and delay on WAN: OSPF calculates cost from bandwidth. IOL defaults to 10 Mbps, making WAN look identical to LAN links. Setting bandwidth 1000 (1 Mbps) now means Project 03 OSPF will calculate realistic costs without redesign.

Step 2.1 — Configure WAN on HQ-RTR1

```
configure terminal


interface Ethernet0/1
description WAN-TO-BR-RTR1-E0/1
ip address 10.0.0.1 255.255.255.252
bandwidth 1000
delay 1000
no shutdown


interface Ethernet0/0.500
description GATEWAY-VLAN500-VOICE
encapsulation dot1Q 500
ip address 10.1.50.1 255.255.255.0


ip route 10.2.100.0 255.255.255.0 10.0.0.2
ip route 10.2.200.0 255.255.255.0 10.0.0.2
ip route 10.2.44.0 255.255.255.0 10.0.0.2
ip route 10.2.50.0 255.255.255.0 10.0.0.2
ip route 10.2.99.0 255.255.255.0 10.0.0.2


end
write memory
```

Step 2.2 — Configure WAN on BR-RTR1

```
configure terminal


interface Ethernet0/1
description WAN-TO-HQ-RTR1-E0/1
```

```
 ip address 10.0.0.2 255.255.255.252
 bandwidth 1000
 delay 1000
 no shutdown

ip route 10.1.100.0 255.255.255.0 10.0.0.1
ip route 10.1.200.0 255.255.255.0 10.0.0.1
ip route 10.1.44.0 255.255.255.0 10.0.0.1
ip route 10.1.50.0 255.255.255.0 10.0.0.1
ip route 10.1.99.0 255.255.255.0 10.0.0.1

end
write memory
```

WHY include 10.1.99.0 route on BR-RTR1: Without this, DHCP relay packets from the branch cannot reach 10.1.99.50 (the DHCP server).

Phase 2 Verification

```
! On HQ-RTR1:
show ip route
ping 10.0.0.2 ! WAN link to BR-RTR1
ping 10.2.99.1 ! Branch management gateway
ping 10.2.99.3 ! BR-ASW1 management
```

```
Connection to 10.1.99.11 closed.
PC-MGMT1:~# ping 10.2.99.3
PING 10.2.99.3 (10.2.99.3): 56 data bytes
64 bytes from 10.2.99.3: seq=737 ttl=253 time=7.757 ms
64 bytes from 10.2.99.3: seq=738 ttl=253 time=7.729 ms
64 bytes from 10.2.99.3: seq=739 ttl=253 time=7.397 ms
64 bytes from 10.2.99.3: seq=740 ttl=253 time=7.463 ms
64 bytes from 10.2.99.3: seq=741 ttl=253 time=6.963 ms
64 bytes from 10.2.99.3: seq=742 ttl=253 time=6.780 ms
64 bytes from 10.2.99.3: seq=743 ttl=253 time=7.065 ms
64 bytes from 10.2.99.3: seq=744 ttl=253 time=7.316 ms
64 bytes from 10.2.99.3: seq=745 ttl=253 time=7.615 ms
64 bytes from 10.2.99.3: seq=746 ttl=253 time=7.132 ms
64 bytes from 10.2.99.3: seq=747 ttl=253 time=7.209 ms
64 bytes from 10.2.99.3: seq=748 ttl=253 time=7.297 ms
64 bytes from 10.2.99.3: seq=749 ttl=253 time=7.724 ms
64 bytes from 10.2.99.3: seq=750 ttl=253 time=7.460 ms
64 bytes from 10.2.99.3: seq=751 ttl=253 time=6.897 ms
64 bytes from 10.2.99.3: seq=752 ttl=253 time=7.473 ms
64 bytes from 10.2.99.3: seq=753 ttl=253 time=8.514 ms
64 bytes from 10.2.99.3: seq=754 ttl=253 time=7.786 ms
64 bytes from 10.2.99.3: seq=755 ttl=253 time=6.703 ms
64 bytes from 10.2.99.3: seq=756 ttl=253 time=7.093 ms
64 bytes from 10.2.99.3: seq=757 ttl=253 time=6.934 ms
64 bytes from 10.2.99.3: seq=758 ttl=253 time=7.664 ms
```

Figure 2.1 — Cross-site WAN ping from HQ-RTR1 confirming static routes working

Phase 3: Centralized DHCP via Relay Agent

Objective: Deploy a Dnsmasq DHCP server at HQ (10.1.99.50) and configure ip helper-address relay on both routers to serve all 8 VLANs across both sites.

> *WHY ip helper-address: DHCP clients broadcast (255.255.255.255). Broadcasts do not cross router boundaries. ip helper-address converts the broadcast to a unicast packet directed to the DHCP server. The relay also inserts the giaddr field — Dnsmasq reads this to select the correct DHCP pool.*

Step 3.1 — Connect HQ-DHCP-DNS to HQ-DSW2

In CML, cable HQ-DHCP-DNS eth0 to HQ-DSW2 Ethernet0/0. Then:

```
! On HQ-DSW2:
configure terminal
interface Ethernet0/0
description TO_HQ-DHCP-DNS
switchport mode access
switchport access vlan 999
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown
end
write memory
```

Step 3.2 — Configure HQ-DHCP-DNS Startup Script

In CML, click HQ-DHCP-DNS → CONFIG tab → paste as Startup Script:

```
#!/bin/sh
hostname hq-dhcp-dns

ip addr add 10.1.99.50/24 dev eth0
ip link set eth0 up

ip route add default via 10.1.99.1

kill -9 $(cat /var/run/dnsmasq.pid) 2>/dev/null
dnsmasq --conf-file=/etc/dnsmasq.conf

echo "=== HQ-DHCP-DNS READY ==="
ip addr show eth0
ip route show
```

Step 3.3 — Configure dnsmasq.conf

In CML, click HQ-DHCP-DNS → CONFIG tab → dnsmasq.conf section. Paste:

```
domain=lab.local
local=/lab.local/
log-dhcp

dhcp-range=set:hq-eng,10.1.100.100,10.1.100.200,255.255.255.0,24h
dhcp-option=tag:hq-eng,option:router,10.1.100.1
dhcp-option=tag:hq-eng,option:dns-server,10.1.99.50

dhcp-range=set:hq-sales,10.1.200.100,10.1.200.200,255.255.255.0,24h
dhcp-option=tag:hq-sales,option:router,10.1.200.1
dhcp-option=tag:hq-sales,option:dns-server,10.1.99.50

dhcp-range=set:hq-guest,10.1.44.100,10.1.44.200,255.255.255.0,24h
dhcp-option=tag:hq-guest,option:router,10.1.44.1
dhcp-option=tag:hq-guest,option:dns-server,10.1.99.50

dhcp-range=set:hq-voice,10.1.50.100,10.1.50.200,255.255.255.0,24h
dhcp-option=tag:hq-voice,option:router,10.1.50.1
dhcp-option=tag:hq-voice,option:dns-server,10.1.99.50

dhcp-range=set:br-eng,10.2.100.100,10.2.100.200,255.255.255.0,24h
dhcp-option=tag:br-eng,option:router,10.2.100.1
dhcp-option=tag:br-eng,option:dns-server,10.1.99.50

dhcp-range=set:br-sales,10.2.200.100,10.2.200.200,255.255.255.0,24h
dhcp-option=tag:br-sales,option:router,10.2.200.1
dhcp-option=tag:br-sales,option:dns-server,10.1.99.50

dhcp-range=set:br-guest,10.2.44.100,10.2.44.200,255.255.255.0,24h
dhcp-option=tag:br-guest,option:router,10.2.44.1
dhcp-option=tag:br-guest,option:dns-server,10.1.99.50

dhcp-range=set:br-voice,10.2.50.100,10.2.50.200,255.255.255.0,24h
dhcp-option=tag:br-voice,option:router,10.2.50.1
```

```
dhcp-option=tag:br-voice,option:dns-server,10.1.99.50


dhcp-host=52:54:00:22:a6:9c,pc-br1,10.2.100.197
dhcp-host=52:54:00:53:da:69,pc-br2,10.2.200.108


address=/hq-rtr1.lab.local/10.1.99.1
address=/hq-dsw1.lab.local/10.1.99.11
address=/hq-dsw2.lab.local/10.1.99.12
address=/hq-asw1.lab.local/10.1.99.13
address=/hq-asw2.lab.local/10.1.99.14
address=/br-rtr1.lab.local/10.2.99.1
address=/br-dsw1.lab.local/10.2.99.2
address=/br-asw1.lab.local/10.2.99.3
address=/hq-dhcp-dns.lab.local/10.1.99.50
address=/pc-br1.lab.local/10.2.100.197
address=/pc-br2.lab.local/10.2.200.108
```

CRITICAL: CRITICAL: Do NOT include bind-interfaces in dnsmasq.conf — causes exit code 30 container crash. Do NOT include interface=eth0 — blocks relayed DHCP packets which arrive as unicast UDP.

Step 3.4 — Configure ip helper-address on both routers

On HQ-RTR1:

```
configure terminal
interface Ethernet0/0.100
ip helper-address 10.1.99.50
interface Ethernet0/0.200
ip helper-address 10.1.99.50
interface Ethernet0/0.300
ip helper-address 10.1.99.50
interface Ethernet0/0.500
ip helper-address 10.1.99.50
end
write memory
```

On BR-RTR1:

```
configure terminal
interface Ethernet0/0.100
ip helper-address 10.1.99.50
```

```
interface Ethernet0/0.200
ip helper-address 10.1.99.50
interface Ethernet0/0.300
ip helper-address 10.1.99.50
interface Ethernet0/0.500
ip helper-address 10.1.99.50
end
write memory
```

WHY helper-address on SUBINTERFACE (not parent E0/0): The relay agent inserts the subinterface IP as giaddr. If you put helper-address on the parent interface, giaddr = 0.0.0.0 and dnsmasq cannot select the right pool.

Phase 3 Verification

Wait 20 seconds after starting HQ-DHCP-DNS, then on PC-BR1:

```
udhcpc -i eth0 -f
```

Expected: Gets an IP in range 10.2.100.100–200 from server 10.1.99.50.

```
! On HQ-DHCP-DNS:
cat /tmp/dnsmasq.log
```

```
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 1:netmask, 3:router, 6:dns-server, 12:hostname,
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 15:domain-name, 28:broadcast, 42:ntp-server
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 next server: 10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  1 option: 53 message-type  2
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 54 server-identifier  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 51 lease-time  1d
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 58 T1  12h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 59 T2  21h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  1 netmask  255.255.255.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 28 broadcast  10.2.100.255
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  6 dns-server  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  3 router  10.2.100.1
Apr  5 20:15:44 dnsmasq-dhcp[1]: no address range available for DHCP request via eth0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 available DHCP range: 10.2.100.100 -- 10.2.100.200
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 vendor class: udhcp 1.37.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 client provides name: localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 DHCPREQUEST(eth0) 10.2.100.197 52:54:00:22:a6:9c
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 tags: br-eng, eth0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 DHCPACK(eth0) 10.2.100.197 52:54:00:22:a6:9c localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 1:netmask, 3:router, 6:dns-server, 12:hostname,
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 15:domain-name, 28:broadcast, 42:ntp-server
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 next server: 10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  1 option: 53 message-type  5
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 54 server-identifier  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 51 lease-time  1d
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 58 T1  12h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 59 T2  21h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  1 netmask  255.255.255.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 28 broadcast  10.2.100.255
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  9 option: 12 hostname  localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  6 dns-server  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  3 router  10.2.100.1
Apr  5 20:15:44 dnsmasq-dhcp[1]: not giving name localhost to the DHCP lease of 10.2.100.197 because the name exis
Apr  5 20:15:45 dnsmasq-dhcp[1]: no address range available for DHCP request via eth0
```

Figure 3.1 — Dnsmasq log showing DHCPDISCOVER and DHCPOFFER with relay giaddr

Phase 4: IPv6 Dual-Stack

Objective: Add IPv6 addressing to the WAN link and VLAN 100 at both sites. Enable SLAAC so endpoints self-configure IPv6 addresses.

WHY /126 on WAN: IPv6 equivalent of /30. Gives exactly 2 usable addresses for point-to-point. The /126 is correct production practice.

WHY 2001:db8:: prefix: IANA-reserved documentation prefix. Safe for labs — never publicly routable.

WHY /64 on VLAN 100: SLAAC requires a /64 prefix. Routers send Router Advertisements with the /64 prefix. End devices use EUI-64 to build a 128-bit global address automatically.

Step 4.1 — Enable IPv6 on HQ-RTR1

```
configure terminal


ipv6 unicast-routing


interface Ethernet0/1
ipv6 address 2001:db8:0:1::1/126
ipv6 enable


interface Ethernet0/0.100
ipv6 address 2001:db8:1:100::1/64
ipv6 enable


ipv6 route 2001:db8:2:100::/64 2001:db8:0:1::2


end
write memory
```

Step 4.2 — Enable IPv6 on BR-RTR1

```
configure terminal
ipv6 unicast-routing


interface Ethernet0/1
ipv6 address 2001:db8:0:1::2/126
ipv6 enable


interface Ethernet0/0.100
ipv6 address 2001:db8:2:100::1/64
ipv6 enable
```

```
ipv6 route 2001:db8:1:100::/64 2001:db8:0:1::1


end
write memory
```

Phase 4 Verification

```
! On HQ-RTR1:
show ipv6 route
ping ipv6 2001:db8:0:1::2


! On PC-BR1 (wait ~10s for RA):
ip -6 addr show eth0
```

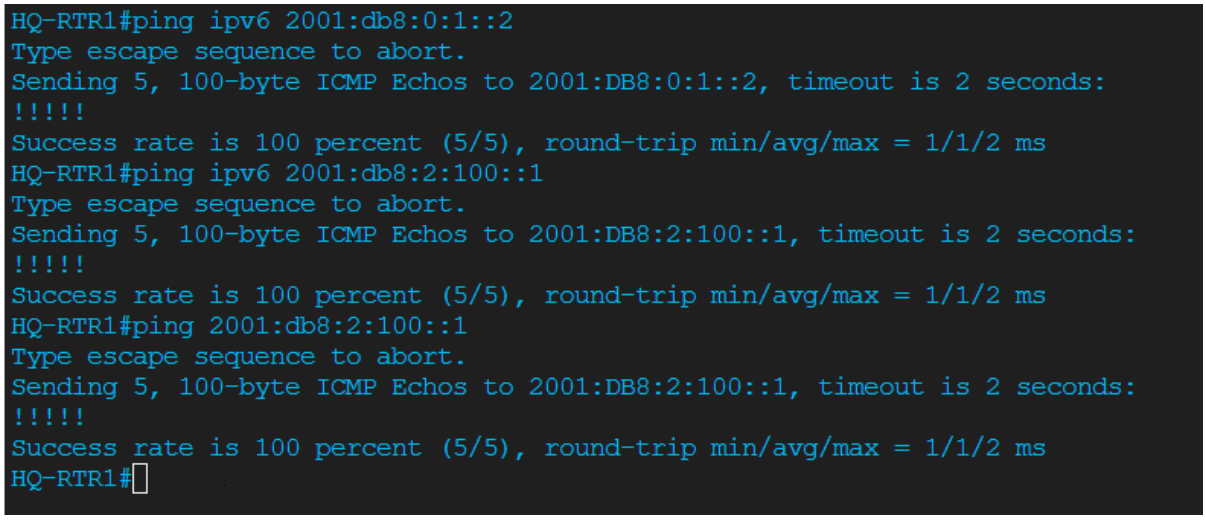

Figure 4.1 — Cross-site IPv6 ping confirming dual-stack WAN connectivity

Phase 5: DNS End-to-End

Objective: Verify and fix DNS resolution so all devices can reach each other by hostname.

Step 5.1 — Re-enable ip domain lookup on routers

Phase 1 hardening applied 'no ip domain lookup' to prevent IOS from DNS-resolving mistyped commands (30-second hang). This same setting blocks legitimate DNS queries from the router.

On HQ-RTR1 and BR-RTR1:

```
configure terminal
ip domain lookup
ip name-server 10.1.99.50
end
write memory
```

Phase 5 Verification

On PC-BR1:

```
nslookup hq-rtr1.lab.local 10.1.99.50
nslookup br-rtr1.lab.local 10.1.99.50
nslookup pc-br2.lab.local 10.1.99.50
```

On BR-RTR1:

```
ping hq-rtr1.lab.local
```

```
** server can't find hq-rtr1.lab.local: NXDOMAIN

pc-br1:~# nslookup br-rtr1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

Name:   br-rtr1.lab.local
Address: 10.2.99.1

** server can't find br-rtr1.lab.local: NXDOMAIN

pc-br1:~# nslookup hq-dhcp-dns.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

Name:   hq-dhcp-dns.lab.local
Address: 10.1.99.50

** server can't find hq-dhcp-dns.lab.local: NXDOMAIN

pc-br1:~# nslookup pc-br1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

** server can't find pc-br1.lab.local: NXDOMAIN

** server can't find pc-br1.lab.local: NXDOMAIN

pc-br1:~# ping -c 3 10.1.99.1
PING 10.1.99.1 (10.1.99.1): 56 data bytes
64 bytes from 10.1.99.1: seq=0 ttl=254 time=4.966 ms
64 bytes from 10.1.99.1: seq=1 ttl=254 time=4.693 ms
64 bytes from 10.1.99.1: seq=2 ttl=254 time=4.672 ms

--- 10.1.99.1 ping statistics ---
```

Figure 5.1 — Successful nslookup on PC-BR1 resolving hq-rtr1.lab.local

Break/Fix Challenge

> **SCENARIO:** SCENARIO: Three simultaneous faults introduced. Two helpdesk tickets arrive: 1. Branch Engineering VLAN 100 — no DHCP 2. HQ Sales VLAN 200 — no DHCP DNS and management connectivity are unaffected. Do NOT read further until you have attempted diagnosis.

Faults Introduced

#	Location	Fault
1	BR-RTR1 E0/0.100	helper-address changed to 10.1.99.99 (wrong server)
2	HQ-RTR1 E0/0.200	helper-address changed to 10.1.99.99 (wrong server)
3	BR-RTR1 routing table	ip route 10.1.99.0 255.255.255.0 Null0 added (black-hole)

Diagnosis Methodology

Layer 1-2: Check physical/VLAN state:

```
show vlan brief
show interfaces trunk
```

Result: Clean — no physical issues.

Layer 3: Check routing table on BR-RTR1:

```
show ip route 10.1.99.0
```

Result: TWO entries — one via 10.0.0.1 (correct) and one via Null0 (black-hole). IOS load-balances. Intermittent ping SUCCESS is a false positive. DHCP relay consistently hits Null0 → silent drop → client timeout.

Layer 7: Check helper-address config:

```
show running-config | section helper
```

Result: ip helper-address 10.1.99.99 on BR-RTR1 E0/0.100 and HQ-RTR1 E0/0.200

Fix — Dependency Order is Critical

```
! Step 1: Remove Null0 FIRST on BR-RTR1
! Fixing helper without this still fails
no ip route 10.1.99.0 255.255.255.0 Null0

! Step 2: Fix BR-RTR1 helper
interface Ethernet0/0.100
ip helper-address 10.1.99.50

! Step 3: Fix HQ-RTR1 helper
interface Ethernet0/0.200
ip helper-address 10.1.99.50
```

```
udhcpc failed to get a DHCP lease
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc failed to get a DHCP lease
^C
pc-br1:~# udhcpc -i eth0
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.2.100.197, server 10.1.99.50
udhcpc: lease of 10.2.100.197 obtained from 10.1.99.50, lease time 86400
pc-br1:~#
```

Figure BF.1 — DHCP restored on PC-BR1 after removing Null0 route and fixing helper

KEY LESSON: KEY LESSON: A successful ping does not prove a clean routing path. A Null0 competitor causes intermittent success that masks the real fault. Always 'show ip route <destination>' before trusting ping. Fix lowest-layer faults first.

Final Verification Checklist

#	Test	Command	Device	Pass Criteria
1	Branch subinterfaces	show ip interface brief	BR-RTR1	E0/0.100-.999 all up/up
2	Branch trunks	show interfaces trunk	BR-DSW1	E0/0+E0/1 trunking, native 1000
3	Branch STP root	show spanning-tree vlan 100	BR-DSW1	This bridge is the root
4	WAN routing	show ip route	HQ-RTR1	5 static routes to 10.2.x.x
5	Cross-site ping	ping 10.2.99.1	HQ-RTR1	5/5 success
6	DHCP relay	udhcpc -i eth0	PC-BR1	IP from 10.1.99.50 in 10.2.100.x
7	Helper-address audit	show run | section helper	Both RTRs	10.1.99.50 on all data subint
8	IPv6 WAN	ping ipv6 2001:db8:0:1::2	HQ-RTR1	5/5 success
9	SLAAC address	ip -6 addr show eth0	PC-BR1	2001:db8:2:100::/64 present
10	DNS resolution	nslookup hq-rtr1.lab.local	PC-BR1	Returns 10.1.99.1

Congratulations — Project 02 Multi-Site Expansion + DHCP is complete. Proceed to Project 03: OSPF Dynamic Routing.